On The Statistical Limits of Self-Improving Agents

Charles L. Wang **Keir Dorchen** **Peter Jin**
Columbia University

Abstract

As systems trend toward superintelligence, a natural modeling premise is that agents can self-improve along every facet of their own design. We formalize this with a five-axis decomposition and a decision layer, separating incentives from learning behavior and analyzing axes in isolation. Our central result identifies and introduces a sharp **utility–learning tension**, the structural conflict in self-modifying systems whereby utility-driven changes that improve immediate or expected performance can also erode the statistical preconditions for reliable learning and generalization. Our findings show that distribution-free guarantees are preserved **iff** the policy-reachable model family is uniformly capacity-bounded; when capacity can grow without limit, utility-rational self-changes can render learnable tasks unlearnable. Under standard assumptions common in practice, these axes reduce to the same capacity criterion, yielding a single boundary for safe self-modification.

1 Introduction

Classical learning theory—from realizable and agnostic PAC to information-theoretic and computational analyses—rests on a tacit premise: the learning mechanism is architecturally invariant. Parameters may adapt, but the agent's update rules, representational scaffolding, topology, computational substrate, and meta-reasoning are treated as fixed. As capabilities trend toward strong open-ended autonomy, however, it is increasingly realistic to assume that advanced agents will self-improve broadly, rewriting not just weights but the very mechanisms by which they learn.

Evidence for this shift already exists. Reinforcement learning and meta-learning instantiate constrained self-change Sutton and Barto 2018; Finn et al. 2017; Rajeswaran et al. 2019; Hospedales et al. 2022, while open-ended pipelines iterate code edits and tools Zhang et al. 2025. Decision-theoretic proposals investigate provably utility-improving modifications Schmidhuber 2005, safety analyses document pathologies Orseau and Ring 2011, and metagoal frameworks aim to stabilize goal evolution Goertzel 2024. What remains underdeveloped is a learning-theoretic account of post-modification behavior: when do seemingly rational self-changes preserve the conditions under which learning is possible, and when do they destroy them? We prove a policy-level learnability boundary: distribution-free PAC guarantees are preserved if and only if the policy-reachable family has uniformly bounded capacity. A simple Two-Gate guardrail, validation margin τ plus capacity cap $K[m]$, keeps trajectories on the safe side and yields a VC-rate oracle inequality.

Our Contributions.

- **Policy boundary with if and only if.** Under standard i.i.d. assumptions, distribution-free PAC learnability is preserved under self-modification if and only if the policy-reachable family has uniformly bounded capacity, measured by VC dimension or an equivalent uniform-convergence notion.

- **Axis reductions.** Architectural and metacognitive edits reduce to induced hypothesis families, and substrate changes matter only via computability or the induced family. Hence the boundary depends solely on the supremum capacity of the reachable family.

- **Two-Gate guardrail.** A computable accept or reject rule, validation improvement by margin τ plus a capacity cap $K[m]$, ensures monotone true-risk progress and an oracle inequality at standard VC rates for the terminal predictor.

- **Axis clarity.** We separate axes that expand the effective hypothesis family from axes that analyze structured subsets within standard models. In particular, the algorithmic axis is a structured subset within standard online learning, not a new model class.

2 Related Work

2.1 Decision-theoretic self-modification and safety

Gödel Machines give a proof-based framework for globally optimal self-modification under a utility function Schmidhuber 2005. Safety analyses document pathologies for self-modifying agents, including reward hacking and self-termination Orseau and Ring 2011. Open-ended empirical systems iterate code and toolchain edits with benchmark gains but without proof obligations Zhang et al. 2025. Proposals for metagoals aim to stabilize or moderate goal evolution during self-change Goertzel 2024. While these frameworks establish decision-theoretic foundations, they do not provide learning-theoretic guarantees about post-modification generalization. We study the learning-theoretic state of the agent after such edits.

2.2 Modern mechanisms for self-improvement

Contemporary machine learning exhibits constrained forms of self-modification across multiple dimensions. Neural architecture search explores architectural topologies through differentiable, evolutionary, and reinforcement approaches Liu et al. 2019; Elsken et al. 2019; Zoph and Le 2017; Real et al. 2019. Automated machine learning systems perform pipeline and hyperparameter search and can trigger optimizer and model-family switches Hutter et al. 2019; Feurer and Hutter 2019; Li et al. 2017. Population-based training simultaneously evolves hyperparameters and weights across a population of models Jaderberg et al. 2017.

Meta-learning adapts optimizers, initializations, and inductive biases across tasks Finn et al. 2017; Rajeswaran et al. 2019; Hospedales et al. 2022. Reinforcement learning and multi-armed bandits provide policies for selecting modifications and exploration strategies Sutton and Barto 2018; Auer et al. 2002; Lai and Robbins 1985; Slivkins 2019. Representation growth through mixture of experts and adapters, and the use of external memory and retrieval, expand the effective function family and computation available at inference Fedus et al. 2022; Houlsby et al. 2019; Hu et al. 2021; Graves et al. 2014, 2016; Lewis et al. 2020; Schick et al. 2023. Continual learning addresses sequential task acquisition while mitigating catastrophic forgetting Kirkpatrick et al. 2017; Parisi et al. 2019; Van de Ven and Tolias 2022.

These mechanisms instantiate partial self-modification, adapting specific components while keeping the learning framework itself fixed. In contrast, true self-modifying agents can rewrite any axis of their design. In our framework, these mechanisms traverse representational, architectural, algorithmic, and metacognitive axes, and we establish when such traversals preserve or destroy learnability.

2.3 Learning theory for adaptive systems

PAC learning provides distribution-free guarantees under a fixed hypothesis class and algorithm Shalev-Shwartz and Ben-David 2014; Mohri et al. 2018; Blumer et al. 1989; Vapnik 1998; Hanneke et al. 2024. Online learning theory establishes regret bounds for adaptive algorithms Shalev-Shwartz 2012; Hazan 2016; Cesa-Bianchi and Lugosi 2006, but assumes the learning mechanism remains fixed. Transformation-invariant learners extend instance equivalence while keeping the learning mechanism fixed Shao et al. 2022. Predictive PAC relaxes data assumptions with a fixed learner Pestov 2010, and iterative improvement within constrained design spaces admits PAC-style analysis Attias et al. 2025.

Information-theoretic approaches bound generalization for adaptive and meta-learners via mutual information Jose and Simeone 2021; Chen et al. 2021; Wen et al. 2025. Stability connects optimization choices to generalization bounds Bousquet and Elisseeff 2002; Hardt et al. 2016. All of these results assume architectural invariance: the hypothesis class, update rule, or computational model is fixed ex ante. We remove this assumption and characterize when self-modification preserves PAC learnability.

2.4 Computability and the substrate

Church–Turing equivalent substrates preserve solvability up to simulation overhead, whereas strictly weaker substrates with finite memory can forfeit learnability of classes that are otherwise PAC-learnable. Stronger-than-Turing models change the problem class under discussion Akbari and Harrison-Trainor 2024. This motivates treating substrate edits separately from architectural or representational changes and clarifies when invariance should be expected.

3 Setup and Five-Axis Decomposition

Data and splits. We study supervised learning under an unknown distribution $\mathcal{D}$ over examples $\langle x, y\rangle \in \mathcal{X} \times \mathcal{Y}$. Let $S \sim \mathcal{D}^m$ denote the training sample and $V \sim \mathcal{D}^{n_v}$ an independent validation sample.

Loss, risk, and utility are distinct objects. Fix a loss $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, 1]$ and define population risk $R[h] = \mathbb{E}_{\langle x,y\rangle\sim\mathcal{D}}[\ell[h[x], y]]$ and empirical risks $\widehat{R}_S$ and $\widehat{R}_V$. Utility u is an agent-internal objective used to decide whether to self-modify. It may depend on proxies of risk, resources, or constraints. We do not assume u equals negative loss unless stated.

Learner state and axes. At time $t \in \mathbb{N}$ the learner state is

$$\ell_t = \langle A_t, H_t, Z_t, F_t, M_t\rangle \in \underbrace{\mathcal{L}}_{\mathcal{A}\times\mathcal{H}\times\mathcal{Z}\times\mathcal{F}\times\mathcal{M}},$$

where A is algorithmic, H is representational, Z is architectural, F is substrate, and M is metacognitive.

Finite evidence and the modification map. Let $\mathcal{E}$ denote the space of finite evidence objects available to the agent, such as a current minibatch, a fixed buffer of past data, a predeclared split of S, or other finite statistics derived from the interaction history. At time t the agent has finite evidence $E_t \in \mathcal{E}$. A possibly stochastic modification map $\Phi : \mathcal{L} \times \mathcal{E} \to \mathcal{L}$ updates the system via

$$\ell_{t+1} = \Phi[\ell_t, E_t].$$

For $X \in \{A, H, Z, F, M\}$ with state space $\mathcal{X}$, we write

$$X_{t+1} = \Phi_X[X_t, E_t, \theta_{X,t}], \qquad \theta_{X,t} \in \Theta_X,$$

where Θ_X indexes admissible edits along axis X.

Environment context. We allow utility to depend on an external environment context Env_t, such as available compute, wall-clock time, or deployment constraints, in addition to finite evidence and internal state.

Decision rule and tie-breaking. A candidate modification at time t is executed if and only if there exists a formal proof in the agent's current calculus that it yields an immediate utility increase:

$$u[\Phi[\ell_t, E_t], \mathrm{Env}_t] \;>\; u[\ell_t, \mathrm{Env}_t].$$

If multiple candidates satisfy this criterion, selection is handled by the metacognitive scheduler M_t, such as first-found proof, maximum certified utility gain, or any fixed tie-break rule. Our theorems are stated in terms of the resulting policy-reachable set induced by this selection rule.

Policy-reachable families. For any axis $X \in \{A, H, Z, F, M\}$, let $\mathcal{X}_{\mathrm{reach}}[u]$ be the set of X states appearing along some trajectory generated by the decision semantics from ℓ_0 under utility u.

Learnability, distribution-free PAC and computable PAC. A hypothesis family $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ is distribution-free PAC learnable if there exists a learner Alg such that for all $\varepsilon, \delta \in]0,1[$, for all distributions $\mathcal{D}$, and for $m \geq m_{\mathcal{H}}[\varepsilon, \delta]$, if $S \sim \mathcal{D}^m$ then with probability at least $1-\delta$ the output $\hat{h} = \mathsf{Alg}[S]$ satisfies

$$R\left[\hat{h}\right] \leq \inf_{h \in \mathcal{H}} R[h] + \varepsilon.$$

When we say learnable in the Turing sense, we additionally require that the learner and the hypotheses it outputs are implementable on a Church–Turing equivalent substrate.

Capacity notion. Our results apply with any uniform capacity notion that yields distribution-free uniform convergence, such as VC-subgraph or pseudodimension. For concreteness, for 0–1 loss we use VC dimension, written $\mathrm{VC}[\cdot]$.

Constants and notation. We use absolute constants $c_1, c_2, \dots$ whose values may differ across lemmas but are fixed within a statement. We use $\tilde{O}[\cdot]$ to hide polylogarithmic factors in $m, n_v, 1/\delta$. Probabilities are over the draws of S and V unless specified.

Axis isolation and substrate scope. Throughout, we analyze one axis at a time while holding others fixed. Under Church–Turing equivalent substrates F, learnability refers to classical PAC in the computable sense. Non Church–Turing cases are treated separately in Section 8.

Data-path integrity. Our PAC statements assume S and V are i.i.d. from $\mathcal{D}$ and independent of each other. Permitted data-path operations are those that preserve i.i.d. draws, such as additional i.i.d. samples, balanced but label-independent subsampling, or predeclared splits. If selection depends on labels or on V, standard importance-weighting or covariate-shift corrections must be used; otherwise guarantees may fail.

Axes and interpretation. Each axis corresponds either to expanding the effective hypothesis family or to analyzing a structured subset within a standard model. Representational and architectural axes expand the effective hypothesis family. The algorithmic axis studies a structured subset within standard online learning. The metacognitive axis filters which edits are evaluated and accepted. The substrate axis affects learnability only through computability or by changing the induced hypothesis family.

1. **Algorithmic.** Update rules, schedules, stopping, and internal randomness. This axis is a structured subset within standard online learning.

2. **Representational.** Changes to the hypothesis class or encoding, such as feature maps, basis expansions, unions, and refinements.

3. **Architectural.** Topology and information flow, including wiring, routing, depth or width, and memory addressing.

4. **Substrate.** Computational model and memory semantics, such as the machine model and memory capacity or discipline.

5. **Metacognitive.** A scheduler that selects and approves modifications on an enabled axis.

Why this decomposition matters. Self-improvement is often discussed as a monolith, but it is not: agents can change *what* they can represent, *how* they search, *how* information flows, *what* compute they have, and *how* they choose among modifications. The five-axis decomposition makes this explicit and, crucially, makes it analyzable. Each axis induces a set of post-modification predictors, and the union of what the agent can reach under its decision rule is the only object that matters for distribution-free guarantees. Once phrased this way, the safety question becomes concrete: does self-improvement keep the reachable family capacity-bounded, or does it allow capacity to drift without limit? The decomposition therefore turns an amorphous concern—agents that rewrite themselves—into a small number of verifiable invariants.

Standing assumptions and scope.

A1 Data $\langle x, y\rangle$ are i.i.d. from fixed $\mathcal{D}$. Training $S \sim \mathcal{D}^m$ and validation $V \sim \mathcal{D}^{n_v}$ are independent.

A2 Loss is bounded, $\ell \in [0, 1]$.

A3 Capacity is any uniform-convergence notion, such as VC, pseudodimension, or VC-subgraph. We instantiate VC where convenient.

A4 When a computable proxy B is used, it upper-bounds capacity, $B[\cdot] \geq \mathrm{cap}[\cdot]$.

A5 Substrate semantics. If the substrate F is Church–Turing equivalent, solvability and learnability are measured in the classical computable sense. Non Church–Turing substrates may alter this and are treated separately in Section 8.

A6 Axis isolation. We analyze one axis at a time while holding the others fixed. Multi-axis edits are discussed later.

A7 Compute scope. We study sample complexity, not runtime, unless limits are intrinsic to F.

4 Representational Self-Modification $\mathcal{M}_H$

Setting, fixed versus modifiable. We analyze representational edits while holding the algorithmic procedure A, architecture Z, substrate F, and metacognitive rule M fixed. At time t the learner has representation H_t and a representational edit is

$$H_{t+1} = \Phi_H[H_t, E_t, \theta_t]\,.$$

Data follow $\langle x, y\rangle \sim \mathcal{D}$ i.i.d. The training set $S \sim \mathcal{D}^m$ and validation set $V \sim \mathcal{D}^{n_v}$ are independent. Loss $\ell \in [0, 1]$. Risk is $R[h]$ and empirical risks are $\widehat{R}_S$ and $\widehat{R}_V$. Capacity is measured by $\mathrm{VC}[\cdot]$. Proofs are deferred to Appendix 12.

Utility assumptions. We use two tiers of assumptions on utility. A utility u is reasonable if it is computable from finite state and evidence, normalized to $[0, 1]$, and is non-decreasing in empirical fit on the active finite evidence. For the destruction results only, we additionally assume a capacity-bonus reasonable utility: u strictly increases with a computable bonus term $g[\mathrm{VC}]$ with $g' > 0$.

Reference family. We work with a fixed capped reference family $\mathcal{G}_{K[m]} \subseteq \mathcal{Y}^{\mathcal{X}}$, satisfying $\mathrm{VC}\big[\mathcal{G}_{K[m]}\big] \leq K[m]$ and fixed ex ante before seeing V.

Policy-reachable family. For fixed u,

$$\mathcal{H}_{\mathrm{reach}}[u] = \{H' :\ \exists t \text{ along a policy-reachable trajectory under } u \text{ with } H_t = H'\}\,.$$

Unbounded representational power. The pair $\langle \mathcal{H}, \Phi_H \rangle$ has unbounded representational power if for every $m \in \mathbb{N}$ there exist H, θ, E with

$$\mathrm{VC}[\Phi_H[H, E, \theta]] \;\geq\; m.$$

Local unbounded representational power holds if for every H there exists an edit that increases VC by at least one and can fit the current finite evidence.

Theorem 1. Policy-level learnability boundary Under Assumptions A1 through A7, distribution-free learnability is preserved under representational self-modification if and only if

$$\sup_{H' \in \mathcal{H}_{\mathrm{reach}}[u]} \mathrm{VC}[H'] < \infty.$$

Sketch. Sufficiency follows because a uniform capacity cap gives uniform convergence on a fixed capped reference family for all steps, and ERM or AERM yields the standard VC rate for the terminal predictor. Necessity follows because if capacities along a reachable subsequence diverge, VC lower bounds preclude any distribution-free sample complexity. Full proof appears in Appendix 12.

Theorem 2. Two-Gate finite-sample safety Given S with $|S| = m$ and independent V with $|V| = n_v$, a candidate edit producing H_{new} is accepted only if

$$\begin{aligned} \text{Validation} \quad & \widehat{R}_V[h_{\mathrm{new}}] \leq \widehat{R}_V[h_{\mathrm{old}}] - [2\varepsilon_V + \tau]\,, \\ \text{Capacity} \quad & h_{\mathrm{new}} \in \mathcal{G}_{K[m]} \;\text{ and }\; \mathrm{VC}\big[\mathcal{G}_{K[m]}\big] \leq K[m]\,, \end{aligned}$$

where ε_V is chosen so that, with probability at least $1 - \delta_V$ over V,

$$\sup_{h \in \mathcal{G}_{K[m]}} \Big| R[h] - \widehat{R}_V[h] \Big| \leq \; \varepsilon_V, \qquad \varepsilon_V \asymp \sqrt{\frac{K[m] + \log[1/\delta_V]}{n_v}}.$$

Then with probability at least $1 - \delta_V - \delta$ over draws of V and S: each accepted edit decreases true risk by at least τ, and

$$R[h_T] \;\leq\; \inf_{h \in \mathcal{G}_{K[m]}} R[h] \;+\; \tilde{O}\left[\sqrt{\frac{K[m] + \log[1/\delta]}{m}}\right].$$

Validation reuse, fixed ex ante. The same validation set V may be reused adaptively across many edits provided the capped reference family $\mathcal{G}_{K[m]}$ is fixed before seeing V and the gate thresholds $K[m]$, ε_V, and τ do not depend on V. If any of these are tuned using V, a fresh split or a reusable holdout is required, as detailed in Appendix 12.

Probability bookkeeping. All oracle inequalities are stated on the intersection of two events: the uniform validation event on $\mathcal{G}_{K[m]}$ with probability at least $1 - \delta_V$ and the training-side uniform convergence event with probability at least $1 - \delta$. By a union bound, the final probability is at least $1 - \delta_V - \delta$ and does not depend on the number of accepted edits, since the bound is uniform over the fixed capped family.

Remark. Under unbounded representational power, utilities that reward empirical fit and even a slight increase in capacity can drive VC unbounded and destroy distribution-free learnability. See Appendix 12.

5 Architectural Self-Modification $\mathcal{M}_Z$

Setting and reduction, fixed versus modifiable. We analyze architectural edits while holding the learning algorithm A, substrate F, and metacognitive rule M fixed. An architecture $Z \in \mathcal{Z}$ induces a hypothesis class $H[Z] \subseteq \mathcal{Y}^{\mathcal{X}}$. At time t, an architectural edit produces

$$Z_{t+1} = \Phi_Z[Z_t, E_t, \vartheta_t]\,, \qquad h_S[Z_{t+1}] \in \arg\min_{h \in H[Z_{t+1}]} \widehat{R}_S[h]\,.$$

Fix a reasonable utility u. Let the policy-reachable architectures and induced classes be

$$\mathcal{Z}_{\text{reach}}[u] = \{Z' : \exists t \text{ on a proof-triggered trajectory from } Z_0 \text{ under } u \text{ with } Z_t = Z'\},$$

and

$$\mathcal{H}^{Z}_{\text{reach}}[u] = \{H[Z] : Z \in \mathcal{Z}_{\text{reach}}[u]\}.$$

Utility realism. The boundary and Two-Gate guarantees depend only on the capacity of the reachable family, not on explicit capacity rewards. Even if u has no bonus term, any policy that permits capacity-increasing edits can cross the boundary unless a cap such as $K[m]$ is enforced. The stronger capacity-bonus variant is used only for destruction results.

Every run of $\mathcal{M}_Z$ corresponds to a run of $\mathcal{M}_H$ over the induced family $\mathcal{H}^{Z}_{\text{reach}}[u]$.

Lemma 3. Architectural to representational reduction For any reasonable u, every proof-triggered trajectory $Z_0 \to Z_1 \to \cdots$ induces a representational trajectory $H[Z_0] \to H[Z_1] \to \cdots$ over a fixed reference family, with ERM or AERM inside each accepted $H[Z_t]$. Consequently $\mathcal{H}^{Z}_{\text{reach}}[u] = \{H[Z] : Z \in \mathcal{Z}_{\text{reach}}[u]\}$. *Proof sketch.* The decision semantics and utility are unchanged by renaming states from Z to the induced $H[Z]$.

Theorem 4. Architectural boundary via induced reachable family For any reasonable u, distribution-free PAC learnability under architectural self-modification is preserved if and only if

$$\sup_{Z \in \mathcal{Z}_{\text{reach}}[u]} \mathrm{VC}[H[Z]] \;\le\; K \;<\; \infty.$$

Equivalently, preservation holds if and only if $\sup_{H' \in \mathcal{H}^{Z}_{\text{reach}}[u]} \mathrm{VC}[H'] \le K$.

Proof. Immediate by reduction to Section 4. Apply the representational boundary theorem, Theorem 1, to the induced set $\mathcal{H}^{Z}_{\text{reach}}[u]$. ☐

Reference family and proxy-cap subfamily. Fix a single parameterized super-family $\mathcal{G} \subseteq \mathcal{Y}^{\mathcal{X}}$ that contains every induced class: $H[Z] \subseteq \mathcal{G}$ for all Z. For $K \in \mathbb{N}$ define the proxy-cap subfamily

$$\mathcal{G}^{\text{proxy}}_{K} = \{h \in \mathcal{G} : \; \exists Z \text{ with } B[Z] \le K \text{ and } h \in H[Z]\},$$

where $B[Z]$ is a computable architectural capacity proxy satisfying $\mathrm{VC}[H[Z]] \le B[Z]$. Since each accepted $H[Z]$ is a subset of the fixed reference subfamily $\mathcal{G}^{\text{proxy}}_{K}$, capacity is bounded by $\mathrm{VC}[\mathcal{G}^{\text{proxy}}_{K}] \le K$.

Corollary 5. Two-Gate safety for architecture Let the capacity gate enforce $H[Z_{\text{new}}] \subseteq \mathcal{G}^{\text{proxy}}_{K[m]}$ with $\mathrm{VC}\Big[\mathcal{G}^{\text{proxy}}_{K[m]}\Big] \le K[m]$, and the validation gate enforce $\widehat{R}_V[h_S[Z_{\text{new}}]] \le \widehat{R}_V[h_S[Z_{\text{old}}]] - [2\varepsilon_V + \tau]$, where ε_V is chosen by a VC bound on $\mathcal{G}^{\text{proxy}}_{K[m]}$. Then the safety and rate conclusions hold as stated by Theorem 2.

Validation reuse. We fix $\mathcal{G}^{\text{proxy}}_{K[m]}$, the schedule $K[m]$, and thresholds ex ante before seeing V. If any choice is tuned on V, we use a fresh split or a reusable holdout mechanism; all theorems apply to the fixed family.

Local architectural edits. We say $\mathcal{M}_Z$ has local unbounded representational power if for every Z there exists a computable edit $\langle \vartheta, E \rangle$ such that $\mathrm{VC}[H[\Phi_Z[Z, E, \vartheta]]] \ge \mathrm{VC}[H[Z]] + 1$ and the new class can interpolate the current finite evidence.

Proposition 6. Robust destruction under local power Assume local unbounded representational power for $\mathcal{M}_Z$. For any capacity-bonus reasonable utility u, there exist a distribution $\mathcal{D}$ and sample size m such that the proof-trigger repeatedly accepts local architectural edits that increase capacity, the induced reachable set $\mathcal{H}^{Z}_{\text{reach}}[u]$ has unbounded VC, and distribution-free PAC learnability fails. *Sketch.* Each local step strictly increases u, so the proof-trigger fires. Iteration yields unbounded VC, then apply Theorem 4.

Proposition 7. Proxy-cap sufficiency If $B[Z] \geq \mathrm{VC}[H[Z]]$ for all Z and the capacity gate enforces $B[Z_{\mathrm{new}}] \leq K[m]$, then the Two-Gate guarantee holds with

$$\mathcal{G}_{K[m]} = \{h \in \mathcal{G} : \ \exists Z \text{ with } B[Z] \leq K[m] \text{ and } h \in H[Z]\},$$

and with the same VC-rate bound.

6 Metacognitive Self-Modification $\mathcal{M}_M$

Setting, fixed versus modifiable. We analyze metacognitive scheduling and filters M while holding representation H, architecture Z, algorithm A, and substrate F fixed. The rule M selects which edits to evaluate and applies acceptance or rejection using finite evidence.

Metacognitive scheduler or filter. A metacognitive rule M can choose which candidate edit to evaluate, when to evaluate, and can randomize; it then accepts or rejects using only finite evidence, such as S, V, a capacity proxy, and edit costs. Let

$$\mathcal{H}^{M,H}_{\mathrm{reach}}[u] = \{H' : \Pr[\exists t \text{ on a proof-triggered trajectory from } H_0 \text{ under } u \text{ filtered by } M \text{ with } H_t = H'] > 0\}.$$

Randomized M is allowed. All guarantees hold almost surely over the internal randomness of M.

Theorem 8. Boundary under metacognitive filtering For any reasonable u and metacognitive filter M, distribution-free PAC learnability is preserved if and only if

$$\sup_{H' \in \mathcal{H}^{M,H}_{\mathrm{reach}}[u]} \mathrm{VC}[H'] \ \leq \ K \ < \ \infty.$$

Proof sketch. Apply the representational boundary theorem to the M filtered family. Necessity follows by the same destruction argument when the supremum is unbounded. Full details appear in Appendix 12.

Two-Gate as metacognition. If M accepts only when the validation margin holds and $\mathrm{VC}[H_{t+1}] \leq K[m]$, then with probability at least $1 - \delta_V - \delta$ each accepted edit reduces true risk by at least τ, and

$$R[h_T] \ \leq \ \inf_{h \in \mathcal{G}_{K[m]}} R[h] \ + \ \tilde{O}\left[\sqrt{\frac{K[m] + \log[1/\delta]}{m}}\right],$$

by Theorem 2.

Restorative metacognition. Suppose the unfiltered reachable family satisfies $\sup_{H' \in \mathcal{H}_{\mathrm{reach}}[u]} \mathrm{VC}[H'] = \infty$. There exists a metacognitive rule M, such as Two-Gate with any computable nondecreasing schedule $K[m]$, such that $\sup_{H' \in \mathcal{H}^{M,H}_{\mathrm{reach}}[u]} \mathrm{VC}[H'] \leq K < \infty$, hence learnability is preserved.

Edit efficiency under margins. Under Two-Gate with margin $\tau > 0$, along any accepted trajectory the number of accepted edits is at most

$$\frac{R[h_0] - R^\star}{\tau}, \qquad R^\star = \inf_{h \in \mathcal{G}_{K[m]}} R[h].$$

Each accepted edit decreases true risk by at least τ, so the bound follows by telescoping.

Scheduling and randomness. Because the deviation bound is uniform over the capped family, the safety and rate guarantees hold for every realized trajectory filtered by M. Scheduling affects efficiency, not the boundary.

7 Algorithmic Self-Modification $\mathcal{M}_A$

Setting, fixed versus modifiable. We analyze algorithmic self-modification while holding the hypothesis class H, architecture Z, substrate F, and metacognitive rule M fixed. The agent may change its update rule, optimizer, or schedule across time, producing

$$A_{t+1} = \Phi_A[A_t, E_t, \vartheta_t] .$$

Given a training set S of size m, the output predictor is $\hat{h} = \mathsf{Alg}[A, S, H] \in H$.

Relation to standard online learning. This axis does not propose a new model class beyond online learning. Any procedure that switches optimizers, such as run SGD then switch to Newton once a condition holds, is still an online learning algorithm because it defines a single map from histories to actions or updates. Our focus is instead a structured subset of online learning algorithms: those that choose among a family of candidate update rules using finite evidence E_t to myopically increase utility. This induces an algorithm-selection effect: when the family of reachable update rules has high statistical capacity, the agent can overfit the choice of update rule to E_t, yielding long-run degradation even when each candidate update rule is itself a valid online learning procedure.

Takeaways. Algorithmic edits cannot cure infinite capacity. On finite capacity, ERM or AERM preserves PAC. When self-modification alters training dynamics beyond ERM assumptions, a simple stability meta-policy based on step-mass controls the generalization gap.

Proposition 9. No algorithmic cure for infinite VC If $\mathrm{VC}[H] = \infty$, then no distribution-free PAC guarantee is possible for any algorithmic procedure. In particular, algorithmic self-modification cannot restore distribution-free learnability.

Proposition 10. Finite VC is sufficient with ERM or AERM If $\mathrm{VC}[H] \le K < \infty$ and the possibly self-modified training procedure is ERM or AERM over H, then for any $\delta \in]0, 1[$, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$,

$$R\left[\hat{h}\right] \;\le\; \inf_{h \in H} R[h] \;+\; \tilde{O}\left[\sqrt{\frac{K + \log[1/\delta]}{m}}\right].$$

Stability meta-policy via step-mass. Assume bounded, Lipschitz, and smooth losses as formalized in Appendix 14, and that training examples are sampled uniformly from S during updates. Let a self-modified training run on H use SGD-like updates with step sizes $\{\eta_t\}_{t=1}^{T}$. Define the step-mass $M_T = \sum_{t=1}^{T} \eta_t$.

Theorem 11. Algorithmic stability via step-mass Under the conditions above, there exists a constant $C > 0$ independent of m such that

$$\mathbb{E}\left[R\left[\hat{h}\right] - \widehat{R}_S\left[\hat{h}\right]\right] \;\le\; \frac{C}{m} \sum_{t=1}^{T} \eta_t \;=\; \frac{C}{m} M_T.$$

A metacognitive rule that caps $M_T \le B[m]$ guarantees $\mathbb{E}[\mathrm{gap}] = \tilde{O}[B[m]\,/m]$. Choosing $B[m] = \tilde{O}[1]$ yields a $\tilde{O}[1/m]$ gap.

Discussion. Proposition 9 says capacity, not optimizer choice, governs distribution-free learnability. Proposition 10 ensures algorithmic edits that continue to output ERM or AERM do not harm PAC guarantees when VC is finite. Theorem 11 offers a simple meta-policy: cap cumulative step-mass to keep the generalization gap small during algorithmic self-modification. Full proofs are in Appendix 14.

8 Substrate Self-Modification $\mathcal{M}_F$

Setting, fixed versus modifiable. We analyze substrate edits while holding the specification of H, Z, and A fixed. Switching substrates changes how these are executed but not which hypotheses are definable nor which utilities are expressible. PAC learnability is classical in the computable sense unless noted.

A substrate edit is

$$F_{t+1} = \Phi_F[F_t, E_t, \varphi_t] \,.$$

Takeaways. First, switching among Church–Turing equivalent substrates preserves classical PAC learnability. Second, downgrading to a strictly weaker substrate, such as finite-state memory, can destroy PAC learnability even for problems learnable pre-switch. Third, stronger-than-Church–Turing substrates do not alter classical PAC guarantees unless they enlarge the induced hypothesis family. If they do, the policy-reachable boundary from Section 4 governs the enlarged family.

Theorem 12. Church–Turing invariance of PAC learnability If F and F' are Church–Turing equivalent, then a problem that is distribution-free PAC learnable when run on F remains distribution-free PAC learnable when run on F', with the same sample complexity up to constant factors. Computation may differ.

Proposition 13. Finite-state downgrade can destroy learnability There exists a binary classification problem that is PAC learnable on a Church–Turing equivalent substrate but becomes not distribution-free PAC learnable after switching to a fixed finite-state substrate with bounded persistent memory, even when H has finite VC dimension as a specification.

Proposition 14. Beyond Church–Turing substrates If a stronger-than-Church–Turing substrate $F^\dagger$ does not enlarge the induced hypothesis family, classical PAC learnability is unchanged. If it enlarges the effective family to $H^\dagger$, the learnability boundary is governed by

$$\sup_{H' \in \mathcal{H}^\dagger_{\mathrm{reach}}[u]} \mathrm{VC}[H'] \,,$$

exactly as in Section 4.

Discussion. Theorem 12 elevates substrate choice out of the classical PAC calculus: Church–Turing equivalent machines affect compute, not sample complexity. Proposition 13 formalizes that collapsing persistent memory imposes an information bottleneck that breaks distribution-free guarantees. Proposition 14 shows that any real change to learnability arises only through the induced hypothesis family. The policy-reachable boundary applies verbatim once that family changes. Full proofs appear in Appendix 15.

9 Outlook

From theory to practice, why capacity bounds matter. Practitioners may object that modern deep learning routinely violates classical worst-case PAC sample complexity through implicit regularization. Why would capacity bounds matter for self-modifying systems? The answer is sequential compounding risk. A single overparameterized model may generalize well, but a self-modifying agent that repeatedly expands capacity across many edits accumulates risk that no implicit bias can reliably rescue.

The Two-Gate policy translates directly into practice: track a capacity proxy $B[\cdot]$, set a schedule $K[m]$ proportional to available data, and reject edits unless validation improves by margin τ. These checks are computationally cheap and can run in real time during self-improvement loops. The alternative to capacity bounds is not trusting implicit regularization. It is accepting that the system has entered a regime where no distribution-free learning guarantee is possible.

Multi-axis modification, the realistic frontier. Real autonomous agents will simultaneously rewrite architectures, swap optimizers, expand tool libraries, and adjust metacognitive policies. Our framework extends naturally: all axes reduce to the same boundary condition on $\sup \mathrm{VC}[\mathcal{H}_{\mathrm{reach}}]$. When an agent modifies multiple axes, the induced hypothesis family is determined by the composition of edits. Learnability requires the joint trajectory remain capacity-bounded. This has three practical implications. First, capacity bounds must be enforced globally, not per-axis. Second, axis interactions create emergent capacity explosions that independent per-axis budgets cannot prevent. Third, metacognitive policies become essential: in multi-axis settings, compounding accelerates capacity growth, making global capacity monitoring the only known mechanism for ex ante safety.

The key open challenge is developing compositional capacity proxies that tractably upper-bound VC for complex compositions. The gap between computable bounds $B[\cdot]$ and true capacity determines how conservative Two-Gate must be.

Towards sustainable self-improvement. AutoML and neural architecture search have achieved success by treating architecture search as unconstrained optimization. Our results suggest a paradigm shift: future self-modifying systems should ask not what maximizes validation accuracy, but what maximizes accuracy subject to capacity remaining PAC learnable for available data. This constraint does not eliminate innovation. It channels self-improvement toward sustainable compounding of gains rather than compounding of risk. For open-ended agents operating over long horizons, the capacity schedule $K[m]$ can grow with accumulating data, enabling unbounded absolute improvement while maintaining learnability. The failure mode is not self-improvement itself, but uncontrolled self-improvement that outruns the data.

10 Conclusion

We have established a sharp learnability boundary for self-modifying agents: distribution-free PAC guarantees are preserved if and only if the policy-reachable hypothesis family has uniformly bounded capacity. This result unifies representational, architectural, algorithmic, metacognitive, and substrate modifications under a single criterion: the supremum VC dimension of states reachable under the agent's utility function. The Two-Gate policy provides a computable guardrail that enforces this boundary through validation margins and capacity caps, yielding oracle inequalities at standard VC rates. Our framework reveals that seemingly rational self-modifications can irreversibly destroy learnability when capacity grows without bound, even as they improve immediate performance. As AI systems gain the capability to rewrite their own learning mechanisms, the choice is between principled capacity-aware self-improvement that preserves generalization guarantees, and unconstrained optimization that enters a regime where learning theory provides no safety assurances.

References

Akbari, S. and Harrison-Trainor, M. (2024). Computable learning of natural hypothesis classes.

Attias, I., Blum, A., Saless, D., Naggita, K., Sharma, D., and Walter, M. (2025). PAC learning with improvements.

Auer, P., Cesa-Bianchi, N., and Fischer, P. (2002). Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47(2-3):235–256.

Blumer, A., Ehrenfeucht, A., Haussler, D., and Warmuth, M. K. (1989). Learnability and the vapnik–chervonenkis dimension. *Journal of the ACM*, 36(4):929–965.

Bousquet, O. and Elisseeff, A. (2002). Stability and generalization. *Journal of Machine Learning Research*, 2:499–526.

Cesa-Bianchi, N. and Lugosi, G. (2006). *Prediction, Learning, and Games*. Cambridge University Press.

Chen, Q., Shui, C., and Marchand, M. (2021). Generalization bounds for meta-learning: An information-theoretic analysis. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 25878–25890.

Elsken, T., Metzen, J. H., and Hutter, F. (2019). Neural architecture search: A survey. *Journal of Machine Learning Research*, 20(55):1–21.

Fedus, W., Zoph, B., and Shazeer, N. (2022). Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39.

Feurer, M. and Hutter, F. (2019). Automl: A survey of the state of the art. In *Automated Machine Learning*, pages 3–33. Springer.

Finn, C., Abbeel, P., and Levine, S. (2017). Model-agnostic meta-learning for fast adaptation of deep networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1126–1135.

Goertzel, B. (2024). Metagoals: Endowing self-modifying AGI systems with goal stability or moderated goal evolution: Toward a formally sound and practical approach.

Graves, A., Wayne, G., and Danihelka, I. (2014). Neural turing machines.

Graves, A., Wayne, G., Reynolds, M., et al. (2016). Hybrid computing using a neural network with dynamic external memory. *Nature*, 538(7626):471–476.

Hanneke, S., Larsen, K. G., and Zhivotovskiy, N. (2024). Revisiting agnostic PAC learning.

Hardt, M., Recht, B., and Singer, Y. (2016). Train faster, generalize better: Stability of stochastic gradient descent. In *International Conference on Machine Learning (ICML)*, pages 1225–1234.

Hazan, E. (2016). *Introduction to Online Convex Optimization*, volume 2. Now Publishers.

Hospedales, T. M., Antoniou, A., Micaelli, P., and Storkey, A. (2022). Meta-learning in neural networks: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(9):5149–5169.

Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. (2019). Parameter-efficient transfer learning for NLP.

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2021). LoRA: Low-rank adaptation of large language models.

Hutter, F., Kotthoff, L., and Vanschoren, J. (2019). *Automated Machine Learning: Methods, Systems, Challenges*. Springer.

Jaderberg, M., Dalibard, V., Osindero, S., et al. (2017). Population based training of neural networks.

Jose, S. T. and Simeone, O. (2021). Information-theoretic generalization bounds for meta-learning and applications. *Entropy*, 23(1):126.

Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., et al. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526.

Lai, T. L. and Robbins, H. (1985). Asymptotically efficient adaptive allocation rules. *Advances in Applied Mathematics*, 6(1):4–22.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Li, L., Jamieson, K., DeSalvo, G., Rostamizadeh, A., and Talwalkar, A. (2017). Hyperband: A novel bandit-based approach to hyperparameter optimization. *The Journal of Machine Learning Research*, 18(1):6765–6816.

Liu, H., Simonyan, K., and Yang, Y. (2019). DARTS: Differentiable architecture search. In *International Conference on Learning Representations (ICLR)*.

Mohri, M., Rostamizadeh, A., and Talwalkar, A. (2018). *Foundations of Machine Learning*. MIT Press, 2nd edition.

Orseau, L. and Ring, M. (2011). Self-modification and mortality in artificial agents. In *Artificial General Intelligence (AGI 2011)*, pages 1–10. Springer.

Parisi, G. I., Kemker, R., Part, J. L., Kanan, C., and Wermter, S. (2019). Continual lifelong learning with neural networks: A review. *Neural Networks*, 113:54–71.

Pestov, V. (2010). Predictive PAC learnability: A paradigm for learning from exchangeable input data.

Rajeswaran, A., Finn, C., Kakade, S., and Levine, S. (2019). Meta-learning with implicit gradients. In *NeurIPS*.

Real, E., Aggarwal, A., Huang, Y., and Le, Q. V. (2019). Regularized evolution for image classifier architecture search. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 4780–4789.

Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Zettlemoyer, L., Cancedda, N., and Scialom, T. (2023). Toolformer: Language models can teach themselves to use tools.

Schmidhuber, J. (2005). Gödel machines: Fully self-referential optimal universal self-improvers. arXiv:cs/0309048.

Shalev-Shwartz, S. (2012). Online learning and online convex optimization. *Foundations and Trends in Machine Learning*, 4(2):107–194.

Shalev-Shwartz, S. and Ben-David, S. (2014). *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press.

Shao, H., Montasser, O., and Blum, A. (2022). A theory of PAC learnability under transformation invariances. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 31904–31917.

Slivkins, A. (2019). Introduction to multi-armed bandits. Foundations and Trends in Machine Learning.

Sutton, R. S. and Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press, 2 edition.

Van de Ven, G. M. and Tolias, A. S. (2022). Three types of incremental learning. *Nature Machine Intelligence*, 4(12):1185–1197.

Vapnik, V. N. (1998). *Statistical Learning Theory*. Wiley.

Wen, W., Gong, T., Dong, Y., Zhang, W., and Liu, Y.-J. (2025). Towards sharper information-theoretic generalization bounds for meta-learning.

Zhang, J., Hu, S., Lu, C., Lange, R., and Clune, J. (2025). Darwin godel machine: Open-ended evolution of self-improving agents.

Zoph, B. and Le, Q. V. (2017). Neural architecture search with reinforcement learning. In *International Conference on Learning Representations*.

Appendix

11 Definitions

Symbol	Definitions
$H(Z)$	Hypothesis class induced by architecture Z
$\mathcal{H}_{\text{reach}}(u)$	Policy-reachable hypothesis family under utility u
$B(\cdot)$	Computable capacity proxy (upper-bounds VC/pseudodim)
$K(m)$	Nondecreasing capacity cap schedule at sample size m
$\mathcal{G}_{K(m)}$	Reference subfamily with cap $\leq K(m)$
τ	Validation margin in Two-Gate
$\widehat{R}_S, \widehat{R}_V$	Empirical risks on train S and validation V
R^*	$\inf_{h \in \mathcal{G}_{K(m)}} R(h)$

Table 1: Notation.

12 Full Proofs for Representational Self-Modification

Data, loss, risks. Samples $(x, y) \sim \mathcal{D}$ i.i.d. Training $S \sim \mathcal{D}^m$ and validation $V \sim \mathcal{D}^{n_v}$ are independent. Loss $\ell \in [0, 1]$; true risk $R(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h(x), y)]$. Empirical risks are $\widehat{R}_S$ and $\widehat{R}_V$ on S and V.

Representational edits and policies. At time t the representation is a hypothesis class $H_t \subseteq \mathcal{Y}^{\mathcal{X}}$. A *representational edit* is $H_{t+1} = \Phi_H(H_t, D_t, \theta_t)$, where D_t is finite evidence (e.g., S, summary stats) and θ_t are edit parameters. Within any accepted class H, the learner outputs an ERM (or AERM) on S:

$$h_S(H) \in \arg\min_{h \in H} \widehat{R}_S(h).$$

The decision rule executes an edit only when an *immediate utility increase is formally provable* from the finite evidence.

Reasonable utilities. A utility u is *reasonable* if it is (i) computable from finite state/evidence and (ii) satisfies:

- **(U1)** Non-decreasing in empirical fit on the active finite evidence (e.g., $1 - \widehat{R}_S$).
- **(U2)** Adds a strictly increasing capacity bonus $g(\mathrm{VC}(H))$ with $g'(k) > 0$ for all k.

We normalize $u \in [0, 1]$ WLOG.

Policy-reachable family. Fix u. Let $\mathcal{H}_{\text{reach}}(u)$ be the set of classes H' for which there exists a time t on *some* proof-triggered trajectory from H_0 under u with $H_t = H'$.

URP and Local-URP. The pair $(\mathcal{H}, \Phi_H)$ has *URP* if for every $m \in \mathbb{N}$ there exist (H, θ, D) with $\mathrm{VC}(\Phi_H(H, D, \theta)) \geq m$. It has *Local-URP* if for every H there is an edit (θ, D) such that $\mathrm{VC}(\Phi_H(H, D, \theta)) \geq \mathrm{VC}(H) + 1$ and the new class can interpolate the current finite evidence (e.g., fit S).

Single capped reference family (indexing control). For each $K \in \mathbb{N}$ let $\mathcal{G}_K \subseteq \mathcal{Y}^{\mathcal{X}}$ be a *reference family* with $\mathrm{VC}(\mathcal{G}_K) \le K$ and assume the capacity gate (defined below) guarantees all accepted H satisfy $H \subseteq \mathcal{G}_K$. This avoids pathologies from taking unions over arbitrarily many distinct classes with the same VC cap.

VC uniform convergence. There exists a universal constant $c > 0$ such that for any class G with $\mathrm{VC}(G) \le K$ and any $\delta \in (0,1)$, with probability $\ge 1-\delta$ over a sample of size n,

$$\sup_{h\in G} \left|R(h) - \widehat{R}(h)\right| \;\le\; c\sqrt{\frac{K + \log(1/\delta)}{n}}. \tag{1}$$

We hide polylogarithmic factors in $\tilde{O}(\cdot)$.

12.1 Sharp policy-level boundary

Theorem (Sharp boundary; restated from Thm. 1). For any reasonable u, distribution-free PAC learnability is preserved under representational self-modification *iff* there exists $K < \infty$ such that

$$\sup_{H'\in\mathcal{H}_{\mathrm{reach}}(u)} \mathrm{VC}(H') \;\le\; K.$$

Proof (sufficiency). Fix u and assume $\sup_{H'\in\mathcal{H}_{\mathrm{reach}}(u)} \mathrm{VC}(H') \le K$. Along any policy-reachable run, all classes satisfy $H_t \subseteq \mathcal{G}_K$ with $\mathrm{VC}(\mathcal{G}_K) \le K$. By (1) applied to $\mathcal{G}_K$ and ERM in H_t,

$$R\big(h_S(H_t)\big) \;\le\; \inf_{h\in H_t} R(h) \;+\; \tilde{O}\Big(\sqrt{\tfrac{K+\log(1/\delta)}{m}}\Big)$$

uniformly for all t, with probability $\ge 1-\delta$ over S. In particular the terminal predictor h_T obeys the same bound, so $m = \tilde{O}\big((K + \log(1/\delta))/\epsilon^2\big)$ suffices for (ϵ, δ)-accuracy. □

Proof (necessity). If $\sup_{H'\in\mathcal{H}_{\mathrm{reach}}(u)} \mathrm{VC}(H') = \infty$, then for each k there exists a reachable $H^{(k)}$ with $\mathrm{VC}(H^{(k)}) \ge k$. Classical VC lower bounds imply any distribution-free learner needs $m = \Omega(k/\epsilon)$ samples for (ϵ, δ)-accuracy (even realizable). Since k is unbounded along reachable trajectories, no uniform PAC guarantee exists. □

12.2 Finite-sample safety of the two-gate policy

Two gates. Given train S ($|S| = m$) and independent validation V ($|V| = n_v$), accept an edit only if:

$$\textbf{(Validation)}\quad \widehat{R}_V(h_{\mathrm{new}}) \;\le\; \widehat{R}_V(h_{\mathrm{old}}) \;-\; (2\varepsilon_V + \tau),$$

$$\textbf{(Capacity)}\quad H_{\mathrm{new}} \;\subseteq\; \mathcal{G}_{K(m)} \qquad \text{with} \quad \mathrm{VC}(\mathcal{G}_{K(m)}) \le K(m),$$

where $K(\cdot)$ is nondecreasing, $\tau \ge 0$ is a margin, and ε_V is chosen so that with probability $\ge 1-\delta_V$ over V,

$$\sup_{h\in\mathcal{G}_{K(m)}} \left|R(h) - \widehat{R}_V(h)\right| \;\le\; \varepsilon_V \quad \text{(e.g., } \varepsilon_V \asymp \sqrt{(K(m) + \log(1/\delta_V))/n_v} \text{ by (1)).}$$

Theorem (Two-gate finite-sample safety; restated from Thm. 2). With probability $\ge 1 - \delta_V - \delta$ over (V, S):
(i) each accepted edit decreases true risk by at least τ (monotone steps); and
(ii) the terminal predictor h_T satisfies the oracle inequality

$$R(h_T) \;\le\; \inf_{h\in\mathcal{G}_{K(m)}} R(h) \;+\; \tilde{O}\Big(\sqrt{\tfrac{K(m)+\log(1/\delta)}{m}}\Big).$$

Proof. (i) On the event $\sup_{h\in\mathcal{G}_{K(m)}} |R(h) - \widehat{R}_V(h)| \le \varepsilon_V$,

$$R(h_{\text{new}}) \;\le\; \widehat{R}_V(h_{\text{new}}) + \varepsilon_V \;\le\; \widehat{R}_V(h_{\text{old}}) - (2\varepsilon_V + \tau) + \varepsilon_V \;\le\; R(h_{\text{old}}) - \tau.$$

(ii) By the capacity gate, $h_T \in \mathcal{G}_{K(m)}$. Apply (1) to $\mathcal{G}_{K(m)}$ on S and ERM in H_T to obtain

$$R(h_T) \;\le\; \inf_{h\in\mathcal{G}_{K(m)}} R(h) \;+\; \tilde{O}\Big(\sqrt{\tfrac{K(m)+\log(1/\delta)}{m}}\Big)$$

with probability $\ge 1-\delta$. Union bound with the validation event yields the stated probability. □

12.3 Destruction under (local) URP

Theorem (Existential destruction under URP). Assume URP. There exists a reasonable utility u and a problem that is PAC-learnable in the baseline class such that the proof-triggered policy executes representational edits that render the problem distribution-free unlearnable after modification.

Proof. Let C be any finite-VC concept class (PAC-learnable without modification). Define a computable utility $u = \alpha\,(1 - \widehat{R}_S(h)) + \beta\, g(\mathrm{VC}(H))$ with $\alpha, \beta > 0$ and strictly increasing g. By URP, for the realized S there exists an edit to $H^\star$ with $\mathrm{VC}(H^\star) \ge |S|$ that interpolates S (shatters S). Then u strictly increases (perfect fit plus larger capacity bonus), which is provable from finite evidence; the proof-trigger executes the edit. With $\mathrm{VC}(H^\star) \ge |S|$ and only $|S|$ samples, standard VC lower bounds show distribution-free PAC learnability fails. □

Theorem (Utility-class robust destruction under Local-URP). Assume Local-URP. Then for *any* reasonable utility u (satisfying (U1)–(U2)), there exist a distribution $\mathcal{D}$ and sample size m such that the proof-trigger repeatedly accepts capacity-increasing local edits, the policy-reachable VC is unbounded, and distribution-free PAC learnability fails.

Proof. Local-URP ensures at each step a computable edit with VC increase by at least 1 that preserves or improves empirical fit on the active evidence. By (U1)–(U2), the utility strictly increases at each such step, hence the proof-trigger fires. Iteration yields unbounded policy-reachable VC, so by the necessity part of the boundary theorem learnability cannot be guaranteed distribution-free. □

13 Full Proofs for Architectural Self-Modification

Assumptions & tools (recap). Data $(x, y) \sim \mathcal{D}$ i.i.d.; training $S \sim \mathcal{D}^m$ and validation $V \sim \mathcal{D}^{n_v}$ are independent. Loss $\ell \in [0, 1]$; true risk $R(h) = \mathbb{E}[\ell(h(x), y)]$; empirical risks $\widehat{R}_S, \widehat{R}_V$ on S, V. Within any accepted class H, the learner outputs ERM $h_S(H) \in \arg\min_{h\in H} \widehat{R}_S(h)$. A utility u is *reasonable* if it is computable from finite evidence and satisfies: **(U1)** non-decreasing in empirical fit on the active evidence (e.g., $1 - \widehat{R}_S$), and **(U2)** strictly increasing in a computable capacity bonus $g(\mathrm{VC}(H))$ with $g'(k) > 0$. We use the standard VC uniform-convergence bound: for a class G with $\mathrm{VC}(G) \le K$ and any $\delta \in (0, 1)$,

$$\sup_{h\in G} \big|R(h) - \widehat{R}(h)\big| \;\le\; c\sqrt{\tfrac{K+\log(1/\delta)}{n}} \qquad \text{(with a universal constant } c > 0\text{)}. \tag{2}$$

Logarithmic factors are absorbed in $\tilde{O}(\cdot)$ as needed.

Architectures induce classes and a reference family. An architecture $Z \in \mathcal{Z}$ induces a hypothesis class $H(Z) \subseteq \mathcal{Y}^{\mathcal{X}}$. We assume a single *reference family* $\mathcal{G} \subseteq \mathcal{Y}^{\mathcal{X}}$ such that $H(Z) \subseteq \mathcal{G}$ for all Z. For each $K \in \mathbb{N}$, let $\mathcal{G}_K \subseteq \mathcal{G}$ be a subfamily with $\mathrm{VC}(\mathcal{G}_K) \le K$. (Concrete choices include $\mathcal{G}$ the realizable functions of a supernet, and K derived from an architectural proxy such as parameter count; see Prop. 13.3.)

Policy reachability in architecture. Fix a reasonable utility u and the proof-triggered decision rule. Define

$$\mathcal{Z}_{\text{reach}}(u) \;=\; \Big\{ Z' :\; \exists t \text{ on some proof-triggered trajectory from } Z_0 \text{ under } u \text{ with } Z_t = Z' \Big\},$$

and the induced family

$$\mathcal{H}^{Z}_{\text{reach}}(u) \;=\; \big\{ H(Z) :\; Z \in \mathcal{Z}_{\text{reach}}(u) \big\} \;\subseteq\; \mathcal{G}.$$

13.1 Reduction lemma: $\mathcal{M}_Z \to \mathcal{M}_H$

Lemma (Architectural-to-representational reduction). Fix a reasonable utility u. Any proof-triggered trajectory $Z_0 \to Z_1 \to \cdots$ in $\mathcal{M}_Z$ induces a trajectory $H(Z_0) \to H(Z_1) \to \cdots$ in $\mathcal{M}_H$ over the family $\mathcal{H}^{Z}_{\text{reach}}(u)$, with predictors $h_S(Z_t) \in \arg\min_{h \in H(Z_t)} \widehat{R}_S(h)$ (ERM).

Proof. By definition, at time t the available hypotheses are exactly $H(Z_t)$. The learner outputs ERM within $H(Z_t)$. The proof-triggered rule accepts an edit iff there exists a formal proof (from finite evidence) that u increases. Because u and the decision semantics are identical whether we name the state by Z_t or by $H(Z_t)$, each accepted architectural edit corresponds to an accepted representational edit on the induced class, and vice versa. Thus the reachable set under u maps to $\mathcal{H}^{Z}_{\text{reach}}(u)$, and the architectural run induces a representational run along the mapped classes.

Corollary (Architectural boundary by reduction). For any reasonable u, distribution-free PAC learnability under architectural self-modification is preserved iff $\sup_{Z \in \mathcal{Z}_{\text{reach}}(u)} \mathrm{VC}(H(Z)) \le K < \infty$.

Proof. Apply the sharp boundary theorem for representation (Appendix 12.1) to the induced set $\mathcal{H}^{Z}_{\text{reach}}(u)$ using Lemma 13.1. See also the main-text statement Thm. 4.

13.2 Local architectural edits and robust destruction

Local-URP^{Z}. We say $\mathcal{M}_Z$ has *Local-URP^{Z}* if for every Z there exists a computable edit (ϑ, D) such that: (i) $\mathrm{VC}\big(H(\Phi_Z(Z, D, \vartheta))\big) \;\ge\; \mathrm{VC}(H(Z)) + 1$; and (ii) the new class $H(\Phi_Z(Z, D, \vartheta))$ can interpolate the active finite evidence (e.g., fit S).

Theorem (Robust destruction under Local-URP^{Z}). Assume Local-URP^{Z}. For any reasonable utility u satisfying (U1)–(U2), there exist a distribution $\mathcal{D}$ and sample size m such that the proof-trigger repeatedly accepts local architectural edits that increase capacity, the induced reachable set $\mathcal{H}^{Z}_{\text{reach}}(u)$ has unbounded VC, and distribution-free PAC learnability fails.

Proof. Fix u. From any Z_t, Local-URP^{Z} guarantees a computable edit (ϑ_t, D_t) producing Z_{t+1} with $\mathrm{VC}(H(Z_{t+1})) \ge \mathrm{VC}(H(Z_t)) + 1$ and perfect fit to the active finite evidence (e.g., S). By (U1) the empirical fit term in u is non-worse; by (U2) the capacity bonus $g(\mathrm{VC})$ strictly increases; the net increase in u is a computable fact from finite evidence. Therefore the proof-trigger fires and the edit is accepted. Iterating yields a reachable sequence with unbounded $\mathrm{VC}(H(Z_t))$. Hence $\sup_{H' \in \mathcal{H}^{Z}_{\text{reach}}(u)} \mathrm{VC}(H') = \infty$, and by the necessity direction of the sharp boundary (Appendix 12.1; cf. Thm. 1) distribution-free PAC learnability cannot be guaranteed.

13.3 Proxy-capacity two-gate guarantee

Computable architectural upper bounds. Suppose there exists a computable function $B : \mathcal{Z} \to \mathbb{N}$ with

$$\mathrm{VC}\big(H(Z)\big) \;\le\; B(Z) \qquad \text{for all } Z \in \mathcal{Z}. \tag{3}$$

(Examples: for ReLU networks, $B(Z) = c\, W(Z) \log W(Z)$ with $W(Z)$ the parameter count.)

Proxy-capped reference subfamily. For $K \in \mathbb{N}$ define

$$\mathcal{G}^{\text{proxy}}_{K} \;:=\; \Big\{ h \in \mathcal{G} :\; \exists Z \in \mathcal{Z} \text{ with } B(Z) \le K \text{ and } h \in H(Z) \Big\}.$$

By (3), $\mathrm{VC}\big(\mathcal{G}^{\text{proxy}}_{K}\big) \le K$.

Proposition (Proxy-cap two-gate oracle inequality). Assume the two-gate policy with capacity gate $B(Z_{\mathrm{new}}) \le K(m)$ and validation gate

$$\widehat{R}_V\big(h_S(Z_{\mathrm{new}})\big) \;\le\; \widehat{R}_V\big(h_S(Z_{\mathrm{old}})\big) \;-\; (2\varepsilon_V + \tau),$$

where ε_V is chosen so that $\sup_{h\in\mathcal{G}^{\mathrm{proxy}}_{K(m)}} |R(h) - \widehat{R}_V(h)| \le \varepsilon_V$ with probability $\ge 1-\delta_V$ (e.g., by (2)). Then with probability $\ge 1-\delta_V-\delta$ over (V,S):

1. (Monotone steps) Each accepted architectural edit satisfies $R\big(h_S(Z_{\mathrm{new}})\big) \;\le\; R\big(h_S(Z_{\mathrm{old}})\big) - \tau$.
2. (Oracle inequality) The terminal predictor $h_S(Z_T)$ obeys

$$R\big(h_S(Z_T)\big) \;\le\; \inf_{h\in\mathcal{G}^{\mathrm{proxy}}_{K(m)}} R(h) \;+\; \tilde{O}\Big(\sqrt{\tfrac{K(m)+\log(1/\delta)}{m}}\Big).$$

Proof. By the capacity gate and the definition of $\mathcal{G}^{\mathrm{proxy}}_{K(m)}$, every accepted class $H(Z_{\mathrm{new}})$ is a subset of $\mathcal{G}^{\mathrm{proxy}}_{K(m)}$ with VC bounded by $K(m)$. The monotone-step claim follows exactly as in the representational two-gate proof: on the high-probability validation event,

$$R(h_{\mathrm{new}}) \;\le\; \widehat{R}_V(h_{\mathrm{new}}) + \varepsilon_V \;\le\; \widehat{R}_V(h_{\mathrm{old}}) - (2\varepsilon_V+\tau) + \varepsilon_V \;\le\; R(h_{\mathrm{old}}) - \tau.$$

For the oracle inequality, apply (2) on S to the capped family $\mathcal{G}^{\mathrm{proxy}}_{K(m)}$ and use ERM in the final accepted class. Union bound the training and validation events to obtain probability $\ge 1-\delta_V-\delta$.

Remark (using a fixed reference subfamily). If the capacity gate is stated directly as $H(Z_{\mathrm{new}}) \subseteq \mathcal{G}_{K(m)}$ with $\mathrm{VC}(\mathcal{G}_{K(m)}) \le K(m)$, then Proposition 13.3 holds verbatim with $\mathcal{G}^{\mathrm{proxy}}_{K(m)}$ replaced by $\mathcal{G}_{K(m)}$.

13.4 Pointers back to representational results

Lemma 13.1 allows Theorem 4 in the main text to be proved by direct appeal to the representational sharp boundary (Appendix 12.1; cf. Thm. 1). Corollary 5 (main text) is an instantiation of the two-gate safety theorem (Appendix 12.2) with the architectural capacity gate $H(Z) \subseteq \mathcal{G}_{K(m)}$.

14 Full Proofs for Algorithmic Self-Modification

Assumptions. Data $(x,y)\sim\mathcal{D}$ i.i.d.; $S\sim\mathcal{D}^m$. Loss $\ell(\cdot;z)$ is bounded in $[0,1]$, L-Lipschitz in the parameter θ (with respect to a norm $\|\cdot\|$), and β-smooth. Gradients are bounded $\|\nabla_\theta \ell(\theta;z)\| \le G$, or we use projection onto a bounded domain of diameter D so iterates remain bounded. The hypothesis class $H = \{x \mapsto f_\theta(x) : \theta\in\Theta\}$ is *fixed* throughout the algorithmic edits. When we invoke ERM/AERM, $\hat{h} \in \arg\min_{h\in H} \widehat{R}_S(h)$ (or an approximate minimizer).

14.1 No algorithmic cure for infinite VC

If $\mathrm{VC}(H)=\infty$, classical VC lower bounds imply that for any learning algorithm (possibly randomized), there exist distributions for which, at any sample size m, the algorithm fails to achieve a universal (ϵ,δ) guarantee (even in the realizable case). Algorithmic self-modification selects among training procedures but does not change H, hence does not change the lower bound. Therefore no distribution-free PAC guarantee is possible. □

14.2 ERM/AERM on finite VC

Assume $\mathrm{VC}(H) \le K < \infty$. Standard uniform-convergence bounds give, with probability $\ge 1-\delta$,

$$\sup_{h\in H} \big|R(h) - \widehat{R}_S(h)\big| \;\le\; c\sqrt{\tfrac{K+\log(1/\delta)}{m}}$$

for a universal constant $c > 0$ (polylogs hidden in $\tilde{O}$). If $\hat{h}$ is ERM/AERM in H, then

$$R(\hat{h}) \;\le\; \widehat{R}_S(\hat{h}) + \tilde{O}\Big(\sqrt{\tfrac{K+\log(1/\delta)}{m}}\Big) \;\le\; \inf_{h\in H} \widehat{R}_S(h) + \tilde{O}\Big(\sqrt{\tfrac{K+\log(1/\delta)}{m}}\Big)$$

$$\le\; \inf_{h\in H} R(h) \;+\; \tilde{O}\Big(\sqrt{\tfrac{K+\log(1/\delta)}{m}}\Big).$$

Thus ERM/AERM preserves the PAC rate on a fixed finite-VC class. □

14.3 Proof of Thm. 11: stability via step-mass

We prove a uniform stability bound for (projected) SGD-like updates and then translate it to an expected generalization bound.

Algorithm and neighboring samples. Let $S = (z_1, \ldots, z_m)$ and $S^{(i)}$ be S with the ith example replaced by an independent copy z_i'. Run the same (possibly self-modified) algorithmic schedule on S and $S^{(i)}$ with shared randomness. Denote parameter sequences $\{\theta_t\}_{t=0}^T$ and $\{\theta_t'\}_{t=0}^T$ with updates

$$\theta_{t+1} \;=\; \Pi(\theta_t - \eta_t g_t)\,, \qquad g_t \in \partial\ell(\theta_t; z_{I_t}),$$

where Π is projection (if used) and I_t is the sampled index at step t (uniform on $[m]$ or any scheme that samples from S); define θ_{t+1}' analogously with $S^{(i)}$ and the *same* I_t sequence.

One-step sensitivity. By nonexpansiveness of projection and β-smoothness with step sizes $\eta_t \le 1/\beta$,

$$\|\theta_{t+1} - \theta_{t+1}'\| \;\le\; \|\theta_t - \theta_t'\| + \eta_t\|g_t - g_t'\|.$$

If $I_t \ne i$ then z_{I_t} is identical in both runs and $\|g_t - g_t'\| \le L\|\theta_t - \theta_t'\|$ (by L-Lipschitzness of gradients under smoothness). If $I_t = i$, the gradients can differ by at most $2G$. Taking conditional expectation over I_t (uniform sampling) yields

$$\mathbb{E}\big[\|\theta_{t+1} - \theta_{t+1}'\| \;\big|\; \theta_t, \theta_t'\big] \;\le\; \Big(1 + \tfrac{L}{m}\eta_t\Big)\|\theta_t - \theta_t'\| \;+\; \tfrac{2G}{m}\eta_t.$$

Iterating from identical initialization gives

$$\mathbb{E}\|\theta_T - \theta_T'\| \;\le\; \tfrac{2G}{m}\sum_{t=1}^T \eta_t \prod_{s=t+1}^T \Big(1 + \tfrac{L}{m}\eta_s\Big) \;\le\; \tfrac{2G}{m}\, e^{(L/m)\sum_s \eta_s} \sum_{t=1}^T \eta_t \;\le\; \tfrac{2Ge^L}{m}\sum_{t=1}^T \eta_t,$$

where we used $\sum_s \eta_s \le M_T$ and $M_T/m \le 1$ for the exponent (or simply absorb $e^{(L/m)M_T}$ into the constant).

From parameter sensitivity to loss stability. By L-Lipschitzness of $\ell(\cdot; z)$ in θ,

$$\big|\ell(\theta_T; z) - \ell(\theta_T'; z)\big| \;\le\; L\,\|\theta_T - \theta_T'\|.$$

Taking expectation over all randomness (sample replacement, SGD sampling, and possibly algorithmic self-mod randomness) yields a uniform stability parameter

$$\epsilon_{\text{stab}} \;:=\; \sup_z \; \mathbb{E}\big[\,|\ell(\theta_T; z) - \ell(\theta_T'; z)|\,\big] \;\le\; \frac{C}{m}\sum_{t=1}^T \eta_t$$

with $C := 2LGe^L$ (or any problem-dependent constant absorbing smoothness/diameter factors).

Generalization gap. By standard stability-to-generalization transfer (uniform stability implies $\mathbb{E}[R(\hat{h}) - \widehat{R}_S(\hat{h})] \leq \epsilon_{\text{stab}}$),

$$\mathbb{E}\big[R(\hat{h}) - \widehat{R}_S(\hat{h})\big] \;\leq\; \frac{C}{m}\sum_{t=1}^{T}\eta_t.$$

This proves the claim. The bound extends to self-modified schedules because only the realized step sizes $\{\eta_t\}$ enter the derivation; any randomness/decision logic is handled by the outer expectation, and the nonexpansive/smoothness arguments hold stepwise. ☐

Meta-policy corollary. If a metacognitive rule enforces $M_T = \sum_t \eta_t \leq B(m)$, then $\mathbb{E}[\text{gap}] \leq (C/m)\,B(m)$. Choosing $B(m) = \tilde{O}(1)$ gives a $\tilde{O}(1/m)$ expected gap; more generally, $B(m) = \tilde{O}(\sqrt{m})$ gives $\tilde{O}(1/\sqrt{m})$, etc.

15 Full Proofs for Substrate Self-Modification

Assumptions. i.i.d. data; loss in $[0,1]$; ERM/AERM as specified. A "substrate" is a computational model hosting the same specification (H, Z, A) unless explicitly stated. "CT-equivalent" means mutually simulable (e.g., universal TM, RAM, λ-calc) with finite simulation overhead independent of m.

15.1 CT-invariance (Thm. 12)

Proof. Let $\mathcal{P}$ be a learning problem that is distribution-freely PAC-learnable on F by algorithm Alg producing $\hat{h}(S) \in H$ with sample complexity $m(\epsilon, \delta)$. Since F' is CT-equivalent to F, there exists a simulator $\mathsf{Sim}_{F' \leftarrow F}$ that, given the code of Alg, simulates it on F' with bounded overhead independent of the sample size m. Running $\mathsf{Sim}_{F' \leftarrow F}(\mathsf{Alg})$ on F' yields the *same* output hypothesis $\hat{h}(S)$ for any dataset S. Hence the *distributional* correctness (and therefore sample complexity) is unchanged. Computation time may increase by a constant/ polynomial factor but PAC definitions are insensitive to runtime. Therefore $\mathcal{P}$ remains distribution-freely PAC-learnable on F' with the same $m(\epsilon, \delta)$ up to constants. ☐ ☐

15.2 Finite-state downgrade impossibility (Prop. 13)

Modeling the downgrade. A "finite-state substrate" has a fixed number N of persistent states (independent of m) available to the learner across training and prediction. Internally, any learning procedure is a transducer whose internal memory is one of N states; weights/parameters cannot grow with m.

Problem and intuition. Consider thresholds on $\mathbb{N}$ (or $\{0,1\}^*$ interpreted as integers): for $k \in \mathbb{N}$, define

$$h_k(x) = ⊮\{x \geq k\}.$$

The class $\mathcal{H}_{\text{thr}} = \{h_k : \; k \in \mathbb{N}\}$ has $\mathrm{VC}(\mathcal{H}_{\text{thr}}) = 1$ (finite) and is PAC-learnable on CT-equivalent substrates (e.g., ERM picks $\hat{k}$ between the largest positive and smallest negative). We show a finite-state substrate cannot distribution-freely PAC-learn $\mathcal{H}_{\text{thr}}$.

Lemma (Pumping-style indistinguishability). For any finite-state learner L with N states and any $m > N$, there exist two training samples S, S' of size m such that: (i) L ends in the *same* internal state on S and S' (with the same output hypothesis), yet (ii) there exists a threshold target h_k and a test point $x^\star$ on which the two training histories require *different* predictions to achieve small risk.

Proof. Process the sorted stream of distinct integers $\{1, 2, \dots\}$; after N distinct "milestones" the finite machine must revisit a state (pigeonhole principle). Construct S and S' that are identical except for the placement of one positive/negative label beyond the Nth milestone so that the same terminal state is reached but the correct threshold differs (choose k between the conflicting milestones). Then a test point $x^\star$ between those milestones is labelled differently by the two Bayes-consistent thresholds. Since the learner outputs the same hypothesis after S and S', it must err on one of the two underlying distributions by a constant amount (bounded away from 0).

Proof of Prop. 13. Fix any finite-state learner and $\delta < 1/4$. For each $m > N$, by Lemma 15.2 we can pick a distribution $\mathcal{D}$ supported on a small interval around the conflicting milestones where the Bayes optimal threshold risk is 0 and the learner's hypothesis (being the same for S and S') incurs error at least $c > 0$ with probability at least $1/2$ over the draw of S. Hence no (ϵ, δ) distribution-free PAC guarantee is possible (take $\epsilon < c/2$). The impossibility holds despite $\mathrm{VC}(\mathcal{H}_{\mathrm{thr}}) = 1$, showing the downgrade (finite persistent memory) destroyed learnability. □ □

Remark (other families). An identical argument works for parities over $\{0,1\}^d$ with unbounded d or for intervals on $\mathbb{N}$, and more generally any family whose Bayes-consistent parameter cannot be encoded in $O(1)$ bits independent of m.

15.3 Beyond-CT substrates (Prop. 14)

Proof. If $F^\dagger$ is stronger than Turing but the *measurable hypothesis family under consideration* remains H, then PAC learnability—defined only in terms of approximation and sample complexity—depends on H and the data distribution, not on computational strength. Thus the sample complexity guarantees are unchanged (computation of ERM/AERM might become easier, but that is outside classical PAC). If, however, the substrate edit enlarges the *effective* family to $H^\dagger$ (e.g., enabling hypercomputable predictors or oracular label access), then learnability is governed by the induced policy-reachable family $\mathcal{H}^\dagger_{\mathrm{reach}}(u)$, and *all* results from §4 apply: preservation iff $\sup_{H' \in \mathcal{H}^\dagger_{\mathrm{reach}}(u)} \mathrm{VC}(H') < \infty$, and destruction otherwise. □ □

16 Gödel Machine Foundations

Schmidhuber's Gödel Machine (Schmidhuber, 2005) provides a theoretical framework for self-modifying agents that make provably optimal self-improvements. We summarize the key components.

16.1 Architecture and Decision Rule

A Gödel Machine consists of an initial problem-solving policy $e(1)$, a proof searcher that tests *proof techniques* (programs generating formal proofs), an axiomatic system $\mathcal{A}$ encoding hardware semantics, initial software $p(1)$, environmental assumptions, and a utility function u, plus a candidate modification program `switchprog`.

The machine executes `switchprog` (which may rewrite any part of the system, including the proof searcher itself) only when it has proven a *target theorem*:

$$u\big[s(t_1) \oplus (\texttt{switchbit}(t_1) = \text{‘1’}), \mathrm{Env}(t_1)\big] \;>\; u\big[s(t_1) \oplus (\texttt{switchbit}(t_1) = \text{‘0’}), \mathrm{Env}(t_1)\big], \tag{4}$$

where u is the utility function, $s(t)$ is system state at time t, and $s(t_1) \oplus (\texttt{switchbit}(t_1) = b)$ denotes the hypothetical state with $\texttt{switchbit} := b$. The left side represents executing the modification; the right side represents continuing to search for alternatives.

16.2 Global Optimality

Theorem (Schmidhuber 2005). *Given any formalizable utility function u and assuming consistency of the axiomatic system $\mathcal{A}$, any self-modification obtained through proof of a target theorem is globally optimal: the utility of executing the present `switchprog` is higher than the utility of waiting for the proof searcher to produce an alternative `switchprog` later.*

16.3 Bias-Optimal Proof Search

The initial proof searcher uses Bias-Optimal Proof Search (BIOPS), allocating time to proof techniques w proportional to their prior probability $P(w)$.

Theorem (Schmidhuber 2005). *If some unknown proof technique w requires at most $f(k)$ steps to produce a proof of difficulty measure k, then BIOPS requires at most $O(f(k))$ steps.*

Relation to our framework. While Gödel Machines establish decision-theoretic optimality through formal proof, our work specializes this framework to provide distribution-free PAC learning guarantees by imposing tractable capacity bounds on the reachable hypothesis family.